



Day 11 Task - Mobile Device Forensics



Trios cyber • 16 Sept (Edited 19 Sept)

100 points | Due 18 Sept

Practical Assignment: Mobile Device Forensics Using Android Studio, ADB & ALEAPP

Dear Students,

For this practical assignment, you are required to examine mobile-device evidence using an authorized Android Virtual Device (AVD).

The objective of this exercise is to understand mobile evidence sources, preservation, acquisition, artifact examination and forensic interpretation.

Use only test data created for this practical. Do not examine another person's mobile device or personal data.

Part 1: Android Studio and AVD Setup

Install Android Studio from the official Android developer website:

— Android Studio: <https://developer.android.com/studio>

Create an Android Virtual Device (AVD) using Android Studio Device Manager.

Recommended environment:

- Android 11 or later
- Standard Android device profile
- x86_64 system image where available

Record:

- Android Studio version
- Device/emulator model
- Android version
- API level
- AVD name
- Date and time
- Windows operating system

Take screenshots showing the AVD and Android version.



Part 2: ADB Setup

ADB (Android Debug Bridge) will be used to communicate with the Android emulator and collect accessible evidence.

Official documentation:

— ADB: <https://developer.android.com/tools/adb>

Verify ADB:

- `adb version`
- `adb devices`

Record:

- ADB version
- Emulator/device ID
- Connection status

Take screenshots of the commands and results.

Part 3: Device State and Preservation

Before creating test activity, document:

- Power state
- Lock/unlock state
- Network connectivity
- Date/time
- Battery level
- Android version
- Storage information
- Emulator/device information

You may use Android Settings and ADB.

Useful commands:

- `adb shell getprop ro.product.model`
- `adb shell getprop ro.build.version.release`
- `adb shell getprop ro.build.version.sdk`
- `adb shell date`
- `adb shell df -h`

Explain the basic principles of mobile evidence preservation.

Take screenshots of the documented device state.



Part 4: Create Controlled Test Activity

Using the authorized Android emulator, create:

- At least 3 test contacts
- At least 5 test messages, where available
- At least 3 call records, where available
- At least 3 browser searches
- At least 3 browser visits
- At least 2 test downloads
- At least 3 test photographs/images
- At least 1 test video
- At least 1 application-based activity
- Location activity, where supported

Use only fictional test information.

Record:

- Activity
- Application
- Date/time
- User/account context
- Expected evidence

Take screenshots of selected activities.

If an activity is not supported by the emulator, record "Not Available."

Part 5: Collect Evidence Using ADB

Create a working directory:

- ``mkdir C:\Mobile_Forensics``

Identify accessible Android files:

- ``adb shell ls -la /sdcard/``
- ``adb shell ls -la /sdcard/DCIM/``
- ``adb shell ls -la /sdcard/Download/``
- ``adb shell ls -la /sdcard/Pictures/``

Collect relevant test files using ``adb pull``.

Example:

- ``adb pull /sdcard/DCIM C:\Mobile_Forensics\DCIM``
- ``adb pull /sdcard/Download C:\Mobile_Forensics\Download``



Record:

- Source path
- Destination path
- Files collected
- Date/time
- ADB command used
- Limitations

Do not modify the original evidence.

Part 6: Hash the Collected Evidence

Use SHA-256 to verify important collected files.

In Ubuntu WSL or another approved command-line environment, run:

- ``sha256sum <filename>``

Record:

- File name
- Hash algorithm
- Hash value
- Date/time

Take a screenshot of the hash output.

Part 7: Install ALEAPP

ALEAPP will be used to parse supported Android forensic artifacts.

Obtain ALEAPP from:

- ALEAPP: <https://github.com/abrignoni/ALEAPP>

Use Ubuntu WSL for the analysis.

Check Python:

- ``python3 --version``

Create a working directory:

- ``mkdir -p ~/mobile_forensics``
- ``cd ~/mobile_forensics``

Install ALEAPP according to its official instructions.



Record:

- Python version
- ALEAPP version
- Installation method
- Ubuntu/WSL environment

Take a screenshot showing the installation and help output.

Part 8: Examine Android Artifacts

Use ALEAPP and other approved tools to examine the collected evidence.

Examine available:

- Messages
- Calls
- Contacts
- Browser history
- Search activity
- Downloads
- Application artifacts
- Photos and videos
- Metadata
- Location/GPS information
- Other accessible Android artifacts

For at least 5 relevant artifacts, record:

- Artifact name
- Source/application
- Path/location, where available
- Date/time
- Relevant account/participant information
- Tool used
- What you observed
- Why it may be relevant

Take screenshots of the relevant findings.

If an artifact is unavailable, state:

"Not Available."

Part 9: Examine Media and Location

Examine at least 3 media artifacts.



Record:

- File name
- File type
- Date/time
- File path
- Device/application information
- GPS/location information, where available
- Metadata
- Observation
- Relevance

Examine available location artifacts and record at least 3 location-related observations where available.

Correlate location information with:

- Photos/videos
- Messages
- Calls
- Browser activity
- Application activity

Do not assume that metadata or GPS information alone proves when or where a person was present.

Part 10: Backup and Recoverable Data

Identify whether an available backup or recoverable data source exists.

Determine:

- What the source represents
- Date/time, where available
- Types of data present
- Whether messages are present
- Whether application data is present
- Whether media is present

Clearly distinguish:

- Live-device data
- Backup data
- Recovered/deleted data

Do not claim that deleted data was recovered unless you actually observe recoverable evidence.

If unavailable, state:

"Not Available."



Part 11: Application or Database Examination

Where accessible, examine an application database or other application artifact.

You may use DB Browser for SQLite:

— DB Browser for SQLite: <https://sqlitebrowser.org/>

Record:

- Database/application
- Source/path
- Tool and version
- Relevant tables or records
- Date/time
- Observations
- Limitations

Work on a copy of the evidence.

Part 12: Acquisition and Evidence Handling

Describe the acquisition method used.

For this practical, the baseline method is:

— ADB-based collection/export of accessible data from the authorized Android emulator

Record:

- Acquisition method
- Tool and version
- Device/environment
- Date/time
- Evidence obtained
- Limitations

Physical extraction, rooting and bypassing Android security are not required.

Part 13: Timeline and Cross-Source Correlation

Create a chronological timeline containing at least 10 events where sufficient information is available.

For each event, record:

- Date/time
- Artifact/source
- Observed activity
- Interpretation

Correlate at least 3 different artifact types.



Examples:

- Message + call record
- Browser history + download
- Photo metadata + location
- Call record + contact
- Browser activity + application activity

Clearly distinguish between:

Direct observation:

What the evidence directly shows.

Inference:

What you interpret from one or more artifacts.

Do not state an inference as a confirmed fact.

Part 14: Limitations

Document any limitations encountered.

Examples:

- Android permissions
- Application sandboxing
- Lack of root access
- Encryption
- Emulator limitations
- Unsupported application
- Missing browser data
- Missing location data
- Backup unavailable
- Deleted data not recoverable
- ALEAPP parsing limitations

If an artifact is unavailable, state "Not Available" and explain the limitation where possible.

Part 15: Document Your Work

Prepare a PDF or DOCX report containing:

1. Student Information
2. Objective
3. Android Studio and AVD information
4. ADB information
5. Device state and preservation
6. Controlled test activity



7. Evidence acquisition
8. Evidence hash
9. Messages, calls and contacts
10. Browser artifacts
11. Application artifacts
12. Photos/videos and metadata
13. Location/GPS artifacts
14. Backup/recovered data
15. Acquisition method and limitations
16. Screenshots
17. Timeline
18. Cross-source correlation
19. Direct observations vs inferences
20. Conclusion

Each screenshot must have a short caption explaining what it demonstrates.

Submission Requirements

Submit ONE file only:

- PDF (.pdf)
- Microsoft Word document (.docx)

Important Notes

Use only your own authorized Android Virtual Device.

Do not examine another person's private mobile data.

Do not bypass security controls.

Do not intentionally delete important data.

Use only test accounts, test contacts, test messages, test files and test media.

Do not introduce malware or intentionally malicious software.

Do not modify the original evidence after collection.

Keep original evidence separate from working copies.

Use the prescribed toolchain:

- Android Studio
- Android Virtual Device
- ADB
- Python 3
- ALEAPP



Additional tools may be used where necessary and must be documented.

If an artifact is unavailable, clearly state "Not Available."

Do not fabricate findings.

Do not claim that deleted data was recovered unless you actually observe recoverable evidence.

Your conclusion must be based on the available evidence. Clearly distinguish direct observations from forensic inferences.

Expected Learning Outcomes

After completing this practical, you should be able to:

1. Identify common Android mobile forensic evidence sources.
2. Set up an Android Virtual Device for forensic testing.
3. Use ADB to collect accessible Android evidence.
4. Understand basic mobile evidence preservation.
5. Examine messages, calls, contacts and application artifacts.
6. Examine browser activity and downloads.
7. Examine photographs, videos and metadata.
8. Examine available location/GPS artifacts.
9. Understand backup and recoverable data.
10. Use ALEAPP to parse supported Android artifacts.
11. Understand logical acquisition and its limitations.
12. Build a mobile activity timeline.
13. Correlate multiple evidence sources.
14. Distinguish direct observations from forensic inferences.
15. Present a defensible mobile forensic reconstruction.

Submission: Upload your completed PDF/DOCX report to this Google Classroom assignment before the specified deadline. Late submissions are allowed only till after 48 hours from the time the task was posted but it may result in deduction of marks.

Please read the instructions carefully before beginning the assignment.

Your work

Handed in late

Day11.pdf

PDF



Unsubmit

